

ISEF Project: Bengaluru — The Living City

We build a **living AI simulation of Bengaluru**, using the city's real road network, neighborhoods, hospitals, flood-prone locations and other available infrastructure data. Instead of treating people as numbers moving through a graph, the city contains thousands of autonomous AI citizens with homes, workplaces, routines and limited information.

Every citizen behaves independently. A student goes to school, an office worker travels to work, a delivery driver moves across the city, and families travel between different locations. When everything is normal, they simply live their simulated lives.

Then something goes wrong.

Heavy rainfall might flood several Bengaluru roads. A citizen encounters water on their route and decides whether to turn around, find another road or continue. Someone may report the flooding. Another citizen may call emergency services. People start posting reports and warning others. Traffic builds as hundreds of agents independently reroute. An ambulance trying to reach a patient can now get trapped in the same congestion.

The emergency-service agents receive calls from citizens and must decide which incidents are most urgent and which ambulance, fire unit or response team to send. Hospitals receive patients and have limited capacity. Traffic-management agents may close roads or change routing recommendations. Utility agents attempt repairs. Government or disaster-response agents receive incomplete reports from across the city and must decide where limited resources should go.

Crucially, **nobody sees the entire city**.

A citizen knows what they can observe nearby. A hospital knows its own condition. Emergency services know the calls they have received. Different agents must communicate to build an understanding of what is actually happening.

This means a disaster can emerge from thousands of individually reasonable decisions.

For example:

Heavy rain → citizens report flooding → everyone reroutes → one alternative road becomes overloaded → ambulance response slows → hospital receives patients late → repair vehicles are delayed → another infrastructure failure occurs → disruption spreads across Bengaluru.

Our main research question becomes:

Can societies of communicating AI agents detect and prevent cascading urban failures before they become city-wide disasters?

We compare several versions of Bengaluru: citizens following simple mathematical rules, citizens represented by autonomous LLM agents, LLM citizens with emergency services but limited communication, and a fully communicating multi-agent city with citizens, hospitals, responders and infrastructure operators.

We then create hundreds or thousands of controlled disruptions: flooding, road closures, traffic accidents, power interruptions, hospital overload and combinations of events. For every experiment we measure response time, congestion, number of affected citizens, access to essential services, hospital load, infrastructure failures, recovery time and the size of the final cascade.

One particularly interesting research question is whether **citizen behaviour itself can create or prevent cascading failures**. Giving everyone the same “best” evacuation or routing advice might actually produce a new traffic jam. Citizen reports might allow emergency services to detect flooding earlier—or misinformation and duplicate reports might waste resources. Agents independently attempting to avoid danger could accidentally make the overall city less resilient.

Another experiment studies **emergent information networks**. Instead of giving the central AI perfect knowledge of Bengaluru, information has to originate from citizens: calls, reports, sensor readings and messages. We can measure how much information is necessary before emergency services can correctly identify where the real crisis is occurring.

The final result is not simply a city simulation. We produce maps showing where cascades repeatedly begin, which citizen behaviours make disasters better or worse, which communication structures give emergency responders the best information, which apparently unimportant roads or facilities become unexpectedly critical, and which small intervention prevents the greatest amount of disruption.

The Exhibition

A large screen shows **Bengaluru alive**.

Thousands of tiny AI citizens move across the real road network. You can watch traffic flow, hospitals operate, emergency vehicles move and agents communicate.

Then:

⚠️ EXTREME RAINFALL — BELLANDUR

Water begins blocking roads.

Citizen agents start reacting.

Messages appear:

“Road ahead looks flooded. Finding another route.”

“Calling emergency services.”

“Traffic has stopped near the underpass.”

An emergency-control AI receives these reports and begins making decisions.

The visitor becomes part of the simulation.

They are given a limited emergency budget and perhaps three actions:

Close a road.

Send an emergency team.

Issue a public warning.

Protect a hospital.

Deploy pumps.

Change recommended routes.

They make their choices.

Then the city runs forward for several simulated hours.

At the end:

YOU vs AI EMERGENCY TEAM vs NO INTERVENTION

The screen shows how many citizens remained connected to essential services, emergency response times, congestion, hospital load and total cascade size.

The visitor can then watch a replay showing the exact chain of events that caused the city to recover—or collapse.

Why Bengaluru?

Bengaluru makes this much more than a generic toy city. We can use real geography and publicly available information about congestion, roads, stormwater drains and flood-prone locations. Bengaluru Traffic Police already publishes congestion information, while open civic datasets contain BBMP stormwater-drain maps and flood-vulnerable locations.

Interestingly, Bengaluru itself has also been developing an AI-based mobility digital-twin project intended to simulate traffic and human driving behaviour before infrastructure changes are made. Our project would be different: rather than focusing only on mobility, we would investigate **multi-agent behaviour, emergency communication and cascading failures across interacting urban systems.**

TL;DR

We create a **miniature living Bengaluru populated by thousands of autonomous AI citizens.**

They commute, react to floods and failures, call emergency services, report problems, change routes and interact with hospitals and city authorities.

We then deliberately disrupt the city and study how thousands of individual decisions combine into city-wide behaviour.

The research asks:

Can communicating AI citizens and emergency-service agents detect and stop cascading urban disasters—and can individual behaviour sometimes unintentionally make those disasters worse?

The booth lets visitors trigger or respond to a Bengaluru disaster and then compete against the AI emergency-response system to see who can save the city better.